

Module 6 Summary



Module 6 Review, Reflect, Consolidate, and Practice

Module 6 focused on feasibility analysis, forecasting, and financial decision-making under uncertainty. Throughout the module, you learned how financial analysts evaluate projects, forecast future outcomes, and communicate evidence-based recommendations to decision-makers.

This section summarises what you have learned and links it explicitly to the course learning outcomes.

What You Learned in Module 6

Financial Feasibility Analysis Supports Strategic Decisions

You learned that financial feasibility analysis helps organisations evaluate whether projects, investments, or strategies are likely to create value.

You learned how to:

- Distinguish fixed costs from variable costs.
- Calculate contribution margin and break-even output.
- Forecast revenues, costs, and profitability.
- Apply Net Present Value (NPV) and Internal Rate of Return (IRR).
- Understand why NPV is generally preferred when project rankings conflict.
- Use sensitivity and scenario analysis to evaluate risk and uncertainty.
- Interpret financial results in a business decision context.

You now understand that:

- Financial outcomes depend heavily on assumptions.

- A project can appear viable under one scenario and risky under another.
- Financial analytics should support strategic decisions, not replace judgment.

This directly supports:

CLO1: Explain key concepts and theoretical foundations of financial analytics and their role in strategic decision-making.

You can now evaluate whether projects are financially viable and explain the implications of risk, uncertainty, and assumptions.

Forecasting Uses Historical Data to Predict Financial Outcomes

You learned that forecasting uses historical information to make informed predictions about future financial and business outcomes.

You identified key time-series patterns, including:

- Trend
- Seasonality
- Cycles
- Random variation

You learned that:

- Different forecasting methods are appropriate for different types of financial data.
- Stable, trending, and seasonal series require different modelling approaches.
- Forecasts are uncertain and should be interpreted probabilistically rather than as exact predictions.

This supports:

CLO2: Apply quantitative and time-series techniques to model risk, return, and financial performance in real-world contexts.

You have now applied time-series reasoning directly to forecasting and financial analysis problems.

Quantitative Forecasting Techniques

You implemented and interpreted several forecasting methods commonly used in financial and business analytics.

Simple Moving Averages

You learned that moving averages smooth short-term fluctuations by averaging recent observations.

You saw that:

- Short moving averages react more quickly to changes.
 - Longer moving averages produce smoother forecasts but respond more slowly.
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Exponential Smoothing

You learned that exponential smoothing gives greater weight to more recent observations.

You learned that:

- A high alpha places more weight on recent data.
 - A low alpha places more weight on older history.
 - Exponential smoothing is useful when recent financial conditions contain more information about the near future.
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Regression Forecasting

You learned how regression models can forecast:

- Trend-only series
- Trend and seasonality together

You learned that:

- Trend models estimate average financial growth or decline over time.
- Seasonal dummy variables capture repeating monthly or quarterly financial patterns.
- Regression forecasting provides interpretable relationships between time and financial performance.

This supports:

CLO2: Apply quantitative and time-series techniques to model risk, return, and financial performance in real-world contexts.

You can now apply forecasting techniques to financial and operational datasets.

Forecasting Supports Financial Planning and Risk Management

You learned that forecasting supports:

- Budgeting

- Revenue planning
- Capacity planning
- Investment decisions
- Staffing decisions
- Inventory management
- Financial risk management

You also learned that:

- Forecast uncertainty must be communicated clearly.
- Analysts should explain assumptions and limitations.
- Financial analytics is valuable only when results can inform decisions.

This supports:

CLO3: Use business analytics tools to develop, evaluate, and communicate predictive insights that support financial problem-solving.

You can now generate forecasts and interpret them in practical financial contexts.

Communicating Financial Insights Clearly

Throughout the module, you created and interpreted financial visualisations and forecasting outputs.

You learned how to:

- Present trends and seasonal patterns visually.
- Compare forecasting methods graphically.
- Explain financial results to technical and non-technical audiences.
- Connect analytical outputs to strategic business recommendations.

This supports:

CLO4: Create and communicate financial insights using effective visual, written, and oral formats suited to both technical and non-technical stakeholders.

You can now communicate financial analytics clearly using both quantitative evidence and business language.

Practice Tasks

Practice Task 1: Break-even analysis

A café sells takeaway lunch boxes.

Selling price per lunch box = \$15

Variable cost per lunch box = \$9

Monthly fixed costs = \$3,600

Tasks:

- Calculate contribution margin per unit.
 - Calculate the break-even quantity.
 - If the café sells 800 lunch boxes in a month, calculate profit or loss.
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Practice Task 2: Cost and revenue forecast

A small online course provider expects:

Price per course = \$120

Courses sold = 1,500

Variable cost per course = \$35

Fixed annual costs = \$80,000

Tasks:

- Calculate total revenue.
 - Calculate total variable cost.
 - Calculate total cost.
 - Calculate profit.
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Practice Task 3: NPV

A project costs \$50,000 today and produces cash inflows of \$18,000 in Year 1, \$20,000 in Year 2, and \$25,000 in Year 3. The discount rate is 8%.

Tasks:

- Calculate the present value of each inflow.
 - Calculate NPV.
 - State whether the project should be accepted.
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Practice Task 4: Sensitivity analysis

A project has:

Revenue = \$300,000

Variable costs = \$180,000

Fixed costs = \$90,000

Tasks:

- Calculate base-case profit.
 - Calculate profit if revenue falls by 10%.
 - Calculate profit if variable costs rise by 10%, with revenue unchanged.
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Practice Task 5: Scenario analysis

A project is assessed under three scenarios:

Scenario	Revenue	Variable Costs	Fixed Costs
Best case	\$500,000	\$260,000	\$120,000
Base case	\$420,000	\$250,000	\$120,000
Worst case	\$350,000	\$240,000	\$120,000

Tasks:

- Calculate profit in each scenario.
 - Identify whether the project remains viable in all scenarios.
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Practice Task 6: IRR reasoning

A project requires an initial outlay of \$40,000 and returns \$15,000 each year for 4 years.

Tasks:

- Explain whether the IRR is likely to be above or below 10%.
 - Explain what would make the project more attractive.
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Practice Task 7: Feasibility memo

Write a 5-sentence feasibility recommendation for a new campus coffee kiosk using these facts:

- Initial investment = \$65,000
 - Expected annual net cash inflow = \$18,000 for 5 years
 - Required return = 9%
 - Sales are sensitive to campus foot traffic
 - The university wants a clear recommendation
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Answers available [here \(https://learn.usc.edu.au/courses/35359/files/3294278?wrap=1\)](https://learn.usc.edu.au/courses/35359/files/3294278?wrap=1).

Workshop

Tasks (<https://learn.usc.edu.au/courses/35359/files/3282930?wrap=1>)
(<https://learn.usc.edu.au/courses/35359/files/3282930?wrap=1>)